

Analog Output Gas Detector Heads



CVVS

Analog output
gas detector
head

- A detector head directly outputs analog signals, which makes it simple to connect to higher level systems.
- Socket type sensors are adopted for easy maintenance.
- Earthquake-resistant vehicle-mountable specification heads are available. (KD-5S)
- KD-8 and PD-8F (N) detector heads are able to indicate concentration by themselves.
- PD-8F has a low flow detection function.

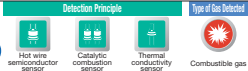
Diffusion type

KD-5S



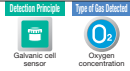
Diffusion type

KD-8



Diffusion type

KD-80



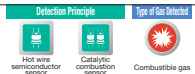
Specifications	
Model	KD-5S
Detection Principle	Hot wire semiconductor or Catalytic combustion
Gas Detected	Combustible gas
Detection Range	As per specifications
Power Source for Sensor	24V DC±10%
Power Consumption	Approx. 5W
Output Signal	4-20mA DC
Maximum Loop Length	1km (with 2mm ² cable)
Sampling Method	Diffusion type
Explosion-Proof Structure	d2G4
Operating Temperature Range	-10°C to 60°C
Applicable Cable	3C shielded
Weight	Approx. 1kg

Specifications	
Model	KD-8
Detection Principle	Hot wire semiconductor, Catalytic combustion, or Thermal conductivity
Gas Detected	Combustible gas
Detection Range	As per specifications
Gas Concentration	-9 to 00%FS LED display (00 flashing at over-range, -9 flashing at below-range)
Indication	As per specifications
Alarm Set Value	As per specifications
Alarm Accuracy	±25% (under identical conditions)
Trouble Indication	LED display (Sensor wire break)
Power Source	100V AC±10% or 24V DC±10%
Power Consumption	Approx. 5VA
Output Signal	4-20mA DC or contact output
Maximum Loop Length	1km (with 2mm ² cable)
Sampling Method	Diffusion type
Explosion-Proof Structure	d2G4
Operating Temperature Range	-10°C to 60°C
Applicable Cable	2C+2C shielded
Dimensions	W224×H335×D153mm(Excluding protrusions)
Weight	Approx. 5kg

Specifications	
Model	KD-80
Detection Principle	Galvanic cell
Gas Detected	Oxygen
Detection Range	As per specifications
Gas Concentration	-9 to 00%FS LED display (00 flashing at over-range, -9 flashing at below-range)
Indication	As per specifications
Alarm Set Value	As per specifications
Alarm Accuracy	± 1vol% (under identical conditions; when the Alarm Set Value is 21vol% or lower)
Trouble Indication	LED display (Sensor failure)
Power Source	100V AC±10% or 24V DC±10%
Power Consumption	Approx. 5VA
Output Signal	4-20mA DC or contact output
Maximum Loop Length	1km (with 2mm ² cable)
Sampling Method	Diffusion type
Explosion-Proof Structure	d2G4
Operating Temperature Range	0°C to 40°C
Applicable Cable	2C+2C shielded
Dimensions	W224×H335×D153mm(Excluding protrusions)
Weight	Approx. 5kg

Suction type

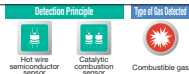
PD-8F



Specifications	
Model	PD-8F (with low flow detection function)
Detection Principle	Hot wire semiconductor or Catalytic combustion
Gas Detected	Combustible gas
Detection Range	As per specifications
Gas Concentration	-9 to 00%FS LED display (00 flashing at over-range, -9 flashing at below-range)
Indication	As per specifications
Alarm Set Value	As per specifications
Alarm Accuracy	±25% (under identical conditions)
Trouble Indication	LED display (Sensor wire break)
Power Source	100V AC±10% or 24V DC±10%
Power Consumption	Approx. 10VA
Output Signal	4-20mA DC or contact output
Maximum Loop Length	1km (with 2mm ² cable)
Sampling Method	Suction type
Explosion-Proof Structure	d2G4
Operating Temperature Range	-10°C to 60°C
Applicable Cable	4C+2C shielded
Dimensions	W224×H335×D153mm(Excluding protrusions)
Weight	Approx. 5kg

Suction type

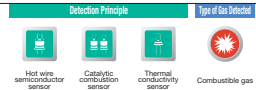
PD-8N



Specifications	
Model	PD-8N (without low flow detection function)
Detection Principle	Hot wire semiconductor or Catalytic combustion
Gas Detected	Combustible gas
Detection Range	As per specifications
Gas Concentration	-9 to 00%FS LED display (00 flashing at over-range, -9 flashing at below-range)
Indication	As per specifications
Alarm Set Value	As per specifications
Alarm Accuracy	±25% (under identical conditions)
Trouble Indication	LED display (Sensor wire break)
Power Source	100V AC±10% or 24V DC±10%
Power Consumption	Approx. 10VA
Output Signal	4-20mA DC or contact output
Maximum Loop Length	1km (with 2mm ² cable)
Sampling Method	Suction type
Explosion-Proof Structure	d2G4
Operating Temperature Range	-10°C to 60°C
Applicable Cable	2C+2C shielded
Dimensions	W224×H452×D174mm(Excluding protrusions)
Weight	Approx. 7kg

Diffusion Type - for Combustible Gas

KD-2A・KD-3A



Model	KD-2A	KD-3A
Detection Principle	Hot wire semiconductor, Catalytic combustion, or Thermal conductivity	
Gas Detected	Combustible gas	
Detection Range	As per specifications	
Power Source for Sensor	Supplied from the indicator unit	
Maximum Loop Length	1km (with 2mm ² cable)	
Sampling Method	Diffusion type	
Explosion-Proof Structure	d2G4	d3aG4/d3cG4
Operating Temperature Range	-10°C to 40°C	
Applicable Cable	4C	
Dimensions	W144×H180×D100mm	
Weight	Approx. 1.2kg	

Diffusion Type - for Combustible Gas

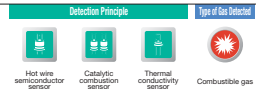
KD-5S・KD-5H



Model	KD-5S	KD-5H
Detection Principle	Hot wire semiconductor or Catalytic combustion	
Gas Detected	Combustible gas	
Detection Range	As per specifications	
Power Source for Sensor	24V DC±10%	
Power Consumption	Approx. 5W	
Output Signal	4-20mA DC	
Maximum Loop Length	1km (with 2mm ² cable)	
Sampling Method	Diffusion type	
Explosion-Proof Structure	d2G4	d3aG4
Operating Temperature Range	-10°C to 60°C	
Applicable Cable	3C shielded	
Dimensions	W141×H192×D92mm	
Weight	Approx. 1kg	

Suction Type - for Combustible Gas

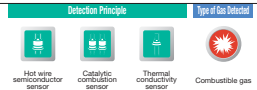
PE-2CC・PE-2DC



Model	PE-2CC	PE-2DC
Detection Principle	Hot wire semiconductor, Catalytic combustion, or Thermal conductivity	
Gas Detected	Combustible gas	
Detection Range	As per specifications	
Power Source for Sensor	Supplied from the indicator unit	
Power Source for Pump	100V AC±10%	24V DC±10%
Maximum Loop Length	1km (with 2mm ² cable)	
Sampling Method	Pump suction type	
Explosion-Proof Structure	d2G4	
Operating Temperature Range	-10°C to 40°C	
Applicable Cable	6C	
Dimensions	W122×H390×D96mm(excluding accessories)	
Weight	Approx. 6.2kg	

Diffusion Type - for Combustible Gas

KD-5A・KD-5B



Model	KD-5A	KD-5B
Detection Principle	Hot wire semiconductor, Catalytic combustion, or Thermal conductivity	
Gas Detected	Combustible gas	
Detection Range	As per specifications	
Power Source for Sensor	Supplied from the indicator unit	
Maximum Loop Length	1km (with 2mm ² cable)	
Sampling Method	Diffusion type	
Explosion-Proof Structure	d3aG4	d2G4
Operating Temperature Range	-10°C to 60°C	
Applicable Cable	3C shielded	
Dimensions	W141×H192×D94mm	
Weight	Approx. 1kg	

Diffusion Type - for Combustible Gas

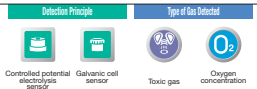
OR-2A



Model	OR-2A
Detection Principle	Orgastor
Gas Detected	Oil, Organic solvent
Power Source for Sensor	Supplied from the indicator unit
Maximum Loop Length	1km (500m when using a Zener barrier; with 2mm ² cable)
Sampling Method	Diffusion type
Explosion-Proof Structure	Intrinsically safe explosion-proof 3nG5 when combined with a Zener barrier
Operating Temperature Range	-10°C to 40°C
Applicable Cable	2C shielded
Dimensions	W40×H35×D87mm

Diffusion Type - for Toxic Gas/Oxygen

KD-5D・KD-5O



Model	KD-5D	KD-5O
Detection Principle	Controlled potential electrolysis	Galvanic cell
Gas Detected	Carbon monoxide	Oxygen
Detection Range	As per specifications	
Power Source for Sensor	Supplied from the indicator unit	
Maximum Loop Length	1km (with 2mm ² cable)	
Sampling Method	Diffusion type	
Explosion-Proof Structure	d2G4	
Operating Temperature Range	-10°C to 60°C	
Applicable Cable	2C shielded	
Dimensions	W141×H192×D94mm	
Weight	Approx. 1kg	

Diffusion Type - for Toxic Gas or Oxygen

KS-2D・KS-2O

Selection Principle

Controlled potential electrolysis sensor

Type of Gas Detected

Galvanic cell sensor

Type of Gas Detected

Toxic gas

Type of Gas Detected

Oxygen concentration



Model	KS-2D	KS-2O
Detection Principle	Controlled potential electrolysis	Galvanic cell
Gas Detected	Toxic gas	Oxygen
Detection Range	As per specifications	
Power Source for Sensor	Supplied from the indicator unit	
Maximum Loop Length	1km (500m when using a Zener barrier; with 2mm ² cable)	
Sampling Method	Diffusion type	
Explosion-Proof Structure	Intrinsically safe explosion-proof 3nG5 when combined with a Zener barrier	
Operating Temperature Range	0°C to 40°C	
Applicable Cable	2C shielded	
Dimensions	W102×H200×D75mm(excluding protrusions)	
Weight	Approx. 1.5kg	

Diffusion Type - for Toxic Gas

KD-2AA・KD-2AB

Selection Principle

Thin film semiconductor sensor

Type of Gas Detected

Toxic gas



Model	KD-2AA	KD-2AB
Detection Principle	Thin film semiconductor	
Gas Detected	Cl ₂	
Detection Range	As per specifications	
Power Source for Sensor	Supplied from the indicator unit	
Maximum Loop Length	1km (with 2mm ² cable)	
Sampling Method	Eductor suction type *1 Diffusion type	
Explosion-Proof Structure	d2G4	Non-explosion-proof
Operating Temperature Range	-10°C to 40°C	
Applicable Cable	4C shielded	
Dimensions	W144×H180×D100mm	
Weight	Approx. 1.2kg	

*1 Separately equipped on a sampling panel

Suction Type - for Toxic Gas

PS-2DPS

Selection Principle

Controlled potential electrolysis sensor

Type of Gas Detected

Toxic gas



Model	PS-2DPS
Detection Principle	Pyrolysis+ Controlled potential electrolysis
Gas Detected	Toxic gas (NF ₃ , CCl ₄)
Detection Range	As per specifications
Power Source for Sensor	Supplied from the indicator unit
Power Source for Pump	100V AC±10%
Maximum Loop Length	1km (with 2mm ² cable)
Sampling Method	Pump suction type
Explosion-Proof Structure	Non-explosion-proof
Operating Temperature Range	0°C to 40°C
Applicable Cable	2C+2C shielded
Dimensions	W300×H350×D100mm(excluding protrusions)
Weight	Approx. 5.6kg

Suction Type - for Toxic Gas

PE-2CZ-NH₃・PE-2DZ-NH₃

Selection Principle

semiconductor sensor

Type of Gas Detected

Toxic gas



Model	PE-2CZ-NH ₃	PE-2DZ-NH ₃
Detection Principle	Semiconductor	
Gas Detected	NH ₃	
Detection Range	As per specifications	
Power Source for Sensor	Supplied from the indicator unit	
Power Source for Pump	100V AC±10%	24V DC±10%
Maximum Loop Length	1km (with 2mm ² cable)	
Sampling Method	Pump suction type	
Explosion-Proof Structure	d2G4	
Operating Temperature Range	-10°C to 40°C	
Applicable Cable	6C	
Dimensions	W122×H390×D96mm(excluding accessories)	
Weight	Approx. 5.6kg	

Diffusion Type - for Toxic Gas

KD-2AS-NH₃

Selection Principle

Semiconductor sensor

Type of Gas Detected

Toxic gas



Model	KD-2AS-NH3
Detection Principle	Semiconductor
Gas Detected	NH ₃
Detection Range	As per specifications
Power Source for Sensor	Supplied from the indicator unit
Maximum Loop Length	1km (with 2mm ² cable)
Sampling Method	Diffusion type
Explosion-Proof Structure	d2G4
Operating Temperature Range	-10°C to 40°C
Applicable Cable	4C shielded
Dimensions	W144×H180×D100mm
Weight	Approx. 1.2kg

Diffusion Type - for Toxic Gas

KCM-3A

Selection Principle

Electrolysis sensor with gel electrolyte

Type of Gas Detected

Toxic gas



Model	KCM-3A
Detection Principle	Electrolysis sensor with gel electrolyte
Gas Detected	COCl ₂ , HCN, others
Detection Range	As per specifications
Power Source for Sensor	Supplied from the indicator unit
Maximum Loop Length	1km (500m when using a Zener barrier; with 2mm ² cable)
Sampling Method	Diffusion type
Explosion-Proof Structure	Intrinsically safe explosion-proof 3nG5 when combined with a Zener barrier
Operating Temperature Range	0°C to 40°C
Applicable Cable	2C shielded
Dimensions	W152×H190×D120mm
Weight	Approx. 1.3kg

Suction Type - for Toxic Gas

PS-2DKP

Selection Principle

Controlled potential electrolysis sensor

Type of Gas Detected

Toxic gas



Model	PS-2DKP
Detection Principle	Controlled potential electrolysis + Light scattering
Gas Detected	SiH ₄ (SiO ₂)
Detection Range	SiH ₄ : 25ppm, SiO ₂ : 10mg/cm ³
Power Source for Sensor	Supplied from the indicator unit
Power Source for Pump	24V DC±10%
Maximum Loop Length	1km (with 2mm ² cable)
Sampling Method	Pump suction type
Explosion-Proof Structure	Non-explosion-proof
Operating Temperature Range	0°C to 40°C
Applicable Cable	2C+2C shielded
Dimensions	W300×H350×D100mm(excluding protrusions)
Weight	Approx. 6.2kg

Suction Type - for Toxic Gas

PS-2CK III

Selection Principle

Controlled potential electrolysis sensor

Type of Gas Detected

Toxic gas



Model	PS-2CKIII
Detection Principle	Pyrolysis ionization
Gas Detected	TEOS and other alkoxide vapors
Detection Range	As per specifications
Power Source for Sensor	Supplied from the indicator unit
Power Source for Pump	100V AC±10%
Maximum Loop Length	1km (with 2mm ² cable)
Sampling Method	Pump suction type
Explosion-Proof Structure	Non-explosion-proof
Operating Temperature Range	0°C to 40°C
Applicable Cable	2C+4C shielded
Dimensions	W300×H350×D100mm(excluding protrusions)
Weight	Approx. 9.3kg

Suction Type - for Toxic Gas

PS-2DP・PS-2DE

Selection Principle

Controlled potential electrolysis sensor

Type of Gas Detected

Toxic gas



Model	PS-2DP	PS-2DE
Detection Principle	Controlled potential electrolysis	
Gas Detected	Toxic gas	
Detection Range	As per specifications	
Power Source for Sensor	Supplied from the indicator unit	
Power Source for Pump	100V AC±10% or 24V DC±10%	
Air Supply		Instrumentation air 0.3-0.7MPa
Maximum Loop Length	1km (with 2mm ² cable)	1km (500m when using a Zener barrier; with 2mm ² cable)
Sampling Method	Pump suction type	Eductor suction type
Explosion-Proof Structure	Non-explosion-proof	Intrinsically safe explosion-proof 3nG5 when combined with a Zener barrier
Operating Temperature Range	0°C to 40°C	
Applicable Cable	2C+2C shielded	2C shielded
Dimensions	W300×H350×D100mm(excluding protrusions)	
Weight	Approx. 5.6kg	

Suction Type - for Oxygen

PS-2OP・PS-2OE

Selection Principle

Galvanic cell sensor

Type of Gas Detected

Oxygen concentration



Model	PS-2OP	PS-2OE
Detection Principle	Galvanic cell	
Gas Detected	Oxygen	
Detection Range	As per specifications	
Power Source for Sensor	Supplied from the indicator unit	
Power Source for Pump	100V AC±10% or 24V DC±10%	
Air Supply		Instrumentation air 0.3-0.7MPa
Maximum Loop Length	1km (with 2mm ² cable)	1km (500m when using a Zener barrier; with 2mm ² cable)
Sampling Method	Pump suction type	Eductor suction type
Explosion-Proof Structure	Non-explosion-proof	Intrinsically safe explosion-proof 3nG5 when combined with a Zener barrier
Operating Temperature Range	0°C to 40°C	
Applicable Cable	2C+2C shielded	2C shielded
Dimensions	W300×H350×D100mm(excluding protrusions)	
Weight	Approx. 5.6kg	

Suction Type - for Toxic Gas

PS-6DKP

Selection Principle

Controlled potential electrolysis sensor

Type of Gas Detected

Toxic gas



Model	PS-6DKP
Detection Principle	Controlled potential electrolysis + Light scattering
Gas Detected	SiH ₄ (SiO ₂)
Detection Range	SiH ₄ : 25ppm, SiO ₂ : 10mg/cm ³
Power Source for Sensor	Supplied from the indicator unit
Power Source for Pump	24V DC±10%
Maximum Loop Length	1km (with 2mm ² cable)
Sampling Method	Pump suction type
Explosion-Proof Structure	Non-explosion-proof
Operating Temperature Range	0°C to 40°C
Applicable Cable	2C+2C shielded
Dimensions	W152×H198×D145mm(excluding protrusions)
Weight	Approx. 6.2kg

Peripherals



Programmer MST-1

Use this programmer (master) to adjust function settings, the zero point, and the span of each gas detector head (slave).



Programmer SST-1

Use this device to calibrate the zero point and the span of a KD-5S gas detector head.



Loader SST-3

Use this loader to adjust settings of an S-500J Receiving unit, including an alarm level and alarm delay time, or to check gas concentration.